

# Incidence-Kernel Bounds for Minimal Counterexamples to the Erdős–Gyárfás Conjecture

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## Abstract

The Erdős–Gyárfás conjecture asserts that every finite graph of minimum degree at least three contains a cycle whose length is a power of two. Carr recently proved that at least  $4/7$  of the vertices in a minimal counterexample would have degree three. Write  $d$  for the number of cubic vertices and  $a$  for the number of vertices of degree at least four. We prove the uniform bound

$$d \geq 2a + 6.$$

Thus every minimal counterexample has at least  $\lceil 2|V(G)|/3 \rceil + 2$  cubic vertices. The key object is a graph on the high-degree vertices obtained by suppressing the cubic vertices with two high-degree neighbours. Four-cycle exclusion and minimality make this incidence kernel simple,  $C_4$ -free, and 2-degenerate; a sharp elementary edge bound and parity then give the result. We further prove, using one sharp extremal theorem and an independently replayable finite certificate, that every such counterexample of order  $n \geq 34$  satisfies

$$d \geq 2a + 7, \quad d \geq \left\lceil \frac{2n + 7}{3} \right\rceil.$$

Thus equality in the first bound can survive only at the seven orders

$$15, 18, 21, 24, 27, 30, 33.$$

## 1 Introduction

The Erdős–Gyárfás conjecture asks whether every finite graph of minimum degree at least three contains a simple cycle of length  $2^k$  for some  $k \geq 2$ . Throughout, graphs are finite, simple, and undirected, and all cycles are simple.

Following Carr [1], call a counterexample *minimal* if it is chosen with minimum possible order and, subject to that, minimum possible size. Carr proved that at least  $4/7$  of its vertices are cubic. His proof records two useful consequences of minimality: the vertices of degree at least four are independent, and the cubic vertices dominate the graph. We reprove those observations and retain more of the resulting incidence structure.

**Theorem 1.** *Let  $G$  be a minimal counterexample to the Erdős–Gyárfás conjecture. Put*

$$D = \{v \in V(G) : d_G(v) = 3\}, \quad A = \{v \in V(G) : d_G(v) \geq 4\},$$

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and write  $d = |D|$ ,  $a = |A|$ . Then

$$d \geq 2a + 6.$$

Consequently, writing  $n = |V(G)|$ ,

$$d \geq \left\lceil \frac{2n+6}{3} \right\rceil = \left\lceil \frac{2n}{3} \right\rceil + 2, \quad a \leq \left\lfloor \frac{n-6}{3} \right\rfloor.$$

## 2 The incidence kernel

We begin with the minimality facts, included to make the argument self-contained.

**Lemma 2.** *Every proper subgraph  $H$  of  $G$  has  $\delta(H) \leq 2$ . Moreover,  $A$  is independent and every vertex of  $G$  has a neighbour in  $D$ .*

*Proof.* If a proper subgraph  $H$  had minimum degree at least three, it would be a counterexample of smaller order, or of the same order and smaller size. If an edge joined two vertices of  $A$ , deleting it would preserve minimum degree at least three. Hence  $A$  is independent. Finally, if a vertex  $v$  had no neighbour in  $D$ , deleting  $v$  would lower each of its neighbours from degree at least four to degree at least three, again producing a smaller counterexample.  $\square$

Every vertex of  $D$  therefore has at most two neighbours in  $A$ . For  $i \in \{0, 1, 2\}$ , let

$$D_i = \{v \in D : d_A(v) = i\}, \quad d_i = |D_i|.$$

Define a graph  $J$  on  $A$  by placing one edge between the two  $A$ -neighbours of each vertex of  $D_2$ .

**Lemma 3.** *The incidence kernel  $J$  is simple,  $C_4$ -free, and 2-degenerate.*

*Proof.* The two  $A$ -neighbours of a vertex in  $D_2$  are distinct. Two different vertices of  $D_2$  cannot have the same neighbour pair, because the two corresponding length-two paths would form a  $C_4$  in  $G$ . Thus  $J$  is simple.

Every simple cycle of length  $\ell$  in  $J$  lifts, by replacing its edges with their corresponding distinct two-edge paths, to a simple cycle of length  $2\ell$  in  $G$ . Hence  $J$ , and every subgraph of  $J$ , has no cycle of power-of-two length. If a subgraph of  $J$  had minimum degree at least three, it would be a counterexample on at most  $a < n$  vertices. This contradicts the order-minimality of  $G$ . Therefore  $J$  is 2-degenerate. In particular it is  $C_4$ -free.  $\square$

We need the following small extremal fact.

**Lemma 4.** *If a simple  $C_4$ -free, 2-degenerate graph has  $r \geq 4$  vertices, then it has at most  $2r - 4$  edges. If  $r \geq 6$ , it has at most  $2r - 5$  edges.*

*Proof.* Every simple 2-degenerate graph on  $r \geq 2$  vertices has at most  $2r - 3$  edges. Equality forces every deletion in a degeneracy ordering to remove a vertex of degree exactly two, eventually leaving a triangle. Reversing the next deletion adds a vertex adjacent to two vertices of that triangle, which creates a  $C_4$ . This proves the first bound.

Suppose now that  $r \geq 6$  and equality  $|E| = 2r - 4$  holds in the first bound. The same deletion argument reduces an equality example to the unique four-vertex, four-edge  $C_4$ -free graph, the paw, and then to the unique five-vertex, six-edge example, two triangles sharing one vertex. Every pair of vertices in the latter graph has a common neighbour. Adding the next degree-two vertex therefore creates a  $C_4$ , a contradiction.  $\square$

### 3 Preliminary abundance bounds

**Lemma 5.** *Every minimal counterexample satisfies  $d \geq 2a + 4$ . If  $a \geq 6$ , then  $d \geq 2a + 6$ .*

*Proof.* First assume  $a \geq 4$ . Since the edges of  $J$  correspond bijectively to the vertices of  $D_2$ , Lemma 4 gives

$$d_2 \leq 2a - 4.$$

Let  $e = e_G(A, D)$  and introduce the nonnegative incidence slacks

$$x = e - 4a, \quad y = 2d - e = d_1 + 2d_0.$$

Since  $e = d_1 + 2d_2$ , we have  $d_1 \geq 8 + x$  and hence  $y \geq 8 + x$ . The exact identity

$$2(d - 2a) = x + y$$

now yields  $d \geq 2a + 4$ .

If  $a \geq 6$ , the second part of Lemma 4 instead gives  $d_2 \leq 2a - 5$ , so  $d_1 \geq 10 + x$  and  $y \geq 10 + x$ . Thus  $d \geq 2a + 5$ . Equality would force  $x = 0$  and  $y = 10$ . But the degree sum inside  $G[D]$  is

$$\sum_{v \in D} d_{G[D]}(v) = 3d - e = d + y,$$

which would be odd when  $d = 2a + 5$  and  $y = 10$ . This contradicts the handshake lemma, and proves  $d \geq 2a + 6$ .

It remains to establish the universal bound for  $a \leq 3$ . If  $a = 0$ , then  $G$  is a nonempty simple cubic graph, so  $d \geq 4$ . If  $a = 1$  and  $d \leq 5$ , the sole  $A$ -vertex has at least four distinct neighbours in  $D$ . For  $d = 4$ , the graph  $G[D]$  is a four-cycle. For  $d = 5$ , parity forces all five vertices of  $D$  to meet  $A$ , so  $G[D]$  is a five-cycle; the  $A$ -vertex together with any two-edge segment of that cycle gives a  $C_4$ . Hence  $d \geq 6$ .

If  $a = 2$ , simplicity and  $C_4$ -freeness give  $d_2 \leq 1$ . Thus

$$8 \leq d_1 + 2d_2 \leq d + 1.$$

The only value below  $2a + 4 = 8$  is  $d = 7$ , which forces  $(d_0, d_1, d_2) = (0, 6, 1)$ . The degree sum in  $G[D]$  is then  $2d_1 + d_2 = 13$ , impossible. Similarly, if  $a = 3$ , then  $d_2 \leq 3$  and

$$12 \leq d_1 + 2d_2 \leq d + 3.$$

The only value below  $2a + 4 = 10$  is  $d = 9$ , forcing  $(d_0, d_1, d_2) = (0, 6, 3)$  and the odd internal degree sum  $2d_1 + d_2 = 15$ . This completes the proof.  $\square$

### 4 Excluding the remaining equality profiles

It remains to rule out  $d = 2a + 4$  and  $d = 2a + 5$  when  $a \leq 5$ . Write  $R = G[D]$  and colour each vertex of  $D_1$  by its unique neighbour in  $A$ .

**Lemma 6.** *If two vertices of  $D_1$  are the endpoints of a two-edge path in  $R$ , then their colours are distinct and have no common neighbour in  $J$ .*

*Proof.* If  $u - z - w$  is such a path and  $u, w$  have the same colour  $\alpha$ , then  $\alpha, u, z, w$  form a four-cycle. Suppose instead that their distinct colours  $\alpha, \beta$  have a common  $J$ -neighbour  $\gamma$ . Let  $p, q \in D_2$  encode the kernel edges  $\alpha\gamma$  and  $\beta\gamma$ . Then

$$\alpha, u, z, w, \beta, q, \gamma, p, \alpha$$

is an eight-cycle. Its vertices are distinct; in particular  $z$  cannot be  $p$  or  $q$ , because vertices of  $D_2$  have degree one in  $R$ .  $\square$

For  $s = d - 2a \in \{4, 5\}$ , the incidence identities take the form

$$x + y = 2s, \quad d_2 = 2a - s + d_0 + x, \quad d_1 = 2s - x - 2d_0. \quad (1)$$

The degree sum  $d + y$  in  $R$  is even. Thus  $x$  is even when  $s = 4$  and odd when  $s = 5$ .

*Completion of the proof of Theorem 1.* Lemma 5 already excludes both values of  $s$  for  $a \geq 6$ . We treat  $a \leq 5$ .

If  $a = 0$ , the case  $s = 4$  is the cubic graph on four vertices, namely  $K_4$ , and the case  $s = 5$  contradicts the handshake lemma. Both are impossible.

Let  $a = 1$ . Since  $d_2 = 0$ , equation (1) leaves, for each  $s$ , either a 2-regular graph  $R$  all of whose vertices have the same colour, or  $d_0 = 2$  and respectively four or five vertices in  $D_1$ . The first alternative contains a two-edge segment closing through the sole  $A$ -vertex to a  $C_4$ . In the second, the two-element  $R$ -neighbourhoods of the  $D_1$  vertices must be pairwise disjoint by Lemma 6; respectively eight elements cannot fit in six vertices, and ten cannot fit in seven.

Let  $a = 2$ . Since  $J$  is simple, equation (1) leaves two profiles for each  $s$ . If  $J$  is empty, then  $R$  is 2-regular on eight or nine coloured vertices. On a cycle of length  $\ell$ , adjacency at distance two gives one odd cycle for odd  $\ell$  and two cycles of length  $\ell/2$  for even  $\ell$ . It is bipartite only when 4 divides  $\ell$ . Avoiding a same-coloured distance-two pair would therefore make every component length divisible by four, forcing a forbidden  $C_4$  or  $C_8$ .

In the other profile,  $J$  is one edge and  $R$  has one degree-three vertex  $z$ , one degree-one vertex  $e \in D_2$ , and all remaining vertices in  $D_1$  of degree two. Its non-cycle component is unicyclic, consisting of a cycle through  $z$  and a pendant path from  $z$  to  $e$ ; any other component would be a coloured cycle and is excluded by the preceding distance-two argument. Since  $e$  is adjacent to both vertices of  $A$ , the pendant path has exactly one coloured internal vertex: zero or at least two would immediately give a  $C_4$ . For  $s = 4$ , the cycle is a six-cycle containing five coloured vertices; three alternating coloured vertices are pairwise at distance two, so two have the same colour and give a  $C_4$ . For  $s = 5$ , write the cycle as

$$z, c_1, c_2, c_3, c_4, c_5, c_6, z.$$

The distance-two constraints on its coloured vertices form the path  $c_2, c_4, c_6, c_1, c_3, c_5$ . If the two colour classes on these six vertices are unbalanced, this path is not properly coloured and gives a  $C_4$ . If they are balanced, proper colouring forces alternation, so  $c_2, c_3$  have the same colour. Their complementary six-edge path around the seven-cycle, together with the two edges through their common  $A$ -neighbour, is a  $C_8$ .

It remains to treat  $a = 3, 4, 5$ . Equations (1), simplicity, and the kernel edge bounds leave exactly the kernels  $P_3$ ,  $K_3$ , the paw, and the five-vertex friendship graph  $F_2$ . For  $s = 4$  their values of  $d_1$  are respectively 8, 6, 8, 8; for  $s = 5$  they are 9, 7, 9, 9.

For  $J = P_3$ , call the colours  $L, M, R$  in path order. Let  $h$  be the number of  $D_1$  vertices with two  $D_1$  neighbours in  $R$ . The two degree-one vertices of  $D_2$  give

$$h \geq 6 \quad (s = 4), \quad h \geq 7 \quad (s = 5).$$

Lemma 6 forces the two neighbour colours of each such vertex to be  $M$  and one of  $L, R$ . There are at most two  $M$ -coloured vertices for  $s = 4$ , and at most three for  $s = 5$ , with total  $R$ -degree at most four and six respectively. Both inequalities are impossible.

For  $J = K_3$ , Lemma 6 says no vertex of  $R$  has two  $D_1$  neighbours. Each of the six or seven vertices of  $D_1$  therefore needs a neighbour outside  $D_1$ . The three degree-one vertices of  $D_2$  supply three incidences, while the sole degree-three vertex of  $D_0$  can supply at most

one without becoming the middle of a forbidden two-edge path. The total capacity four is insufficient.

For the paw, the four degree-one vertices of  $D_2$  imply  $h \geq 4$  for  $s = 4$  and  $h \geq 5$  for  $s = 5$ . The only distinct colour pair without a common  $J$ -neighbour is the degree-three centre and the leaf. Thus every counted vertex must be incident with a centre-coloured vertex. There are respectively at most one and two such vertices, of total  $R$ -degree at most two and four, again a contradiction.

Finally, in  $F_2$  any two distinct colours occurring on  $D_1$  have a common  $J$ -neighbour, including the shared centre if it receives the one extra colour in the  $s = 5$  profile. Hence no vertex of  $R$  has two  $D_1$  neighbours. All eight or nine degree-two vertices of  $D_1$  require a neighbour in  $D_2$ , but the six degree-one vertices of  $D_2$  supply only six incidences.

We have excluded  $s = 4, 5$  in every remaining case. Together with Lemma 5, this proves  $d \geq 2a + 6$ . Substituting  $a = n - d$  gives  $3d \geq 2n + 6$  and  $3a \leq n - 6$ , which are exactly the stated integer bounds.  $\square$

## 5 Excluding the top equality layer

We now sharpen Theorem 1 outside a finite window. This section uses the following result of Györi, Li, Salia, Tompkins, Varga and Zhu [3]: every  $r$ -vertex graph with more than

$$\left\lfloor \frac{19(r-1)}{12} \right\rfloor$$

edges contains a cycle whose length is divisible by four.

Suppose temporarily that equality holds in Theorem 1, so that

$$d = 2a + 6. \tag{2}$$

Put

$$q = 2a - |E(J)| = 2a - d_2.$$

Using the incidence slacks  $x, y$  from Lemma 5, direct substitution gives

$$d_2 = 2a - 6 + x + d_0, \quad q = 6 - x - d_0, \quad d_1 = 12 - x - 2d_0. \tag{3}$$

When  $a \geq 6$ , Lemma 4 gives  $q \geq 5$ . Moreover,  $d$  is even, and the degree sum  $d + y$  inside  $G[D]$  shows that  $y$ , and hence  $x$ , is even. Therefore exactly two profiles remain:

$$q = 5 : \quad x = 0, \quad d_0 = 1, \quad d_1 = 10, \tag{4}$$

$$q = 6 : \quad x = 0, \quad d_0 = 0, \quad d_1 = 12. \tag{5}$$

In both profiles every vertex of  $A$  has degree four, and the number of  $D_1$  vertices whose unique neighbour in  $A$  is  $v$  equals

$$c(v) = 4 - d_J(v). \tag{6}$$

For completeness, the extremal theorem makes both profiles finite. A dyadic-cycle-free, 2-degenerate graph of deficit five,  $|E(H)| = 2|V(H)| - 5$ , is connected once its order is at least five. From order seven onward it has no degree-one vertex; 2-degeneracy therefore supplies a degree-two vertex whose deletion preserves the deficit. Thus an example of order at least nine would reduce to order nine, where it has thirteen edges, exceeding  $\lfloor 19(9-1)/12 \rfloor = 12$ . The resulting cycle divisible by four has length four or eight, a contradiction. Hence the profile (4) has  $a \leq 8$ .

The same deficit addition shows that a disconnected deficit-six graph has order at most six. Once deficit five is excluded from order nine onward, a deficit-six graph of order at least eleven has no degree-one vertex and again deletes through degree two. At order eleven its sixteen edges exceed  $\lfloor 19(11 - 1)/12 \rfloor = 15$ . Hence the profile (5) has  $a \leq 10$ . It follows already that equality (2) is impossible for  $a \geq 11$ .

It remains to exclude  $a = 10$  in the deficit-six profile. Here  $J$  is a connected ten-vertex, fourteen-edge graph with

$$2 \leq \delta(J) \leq \Delta(J) \leq 4,$$

which is 2-degenerate and has no  $C_4$  or  $C_8$ . Connectedness follows by adding the component deficits; a degree-one vertex would delete to the already excluded nine-vertex deficit-five case. Exact canonical generation leaves precisely the following four graph6 records:

```
I?`D@`keO
I?`D@pScg
I?`DCdcMG
I?`@dEWR_
```

All four have degree sequence 2, 2, 2, 2, 3, 3, 3, 3, 4, 4.

We describe the independently replayable finite obstruction. Let  $R = G[D]$ . Each of the fourteen vertices in  $D_2$  has degree one in  $R$ , each of the twelve vertices in  $D_1$  has degree two, and  $D_0$  is empty. Thus  $R$  is a disjoint union of exactly seven paths with  $D_2$  endpoints, together with cycles consisting only of  $D_1$  vertices. If  $\lambda_1, \dots, \lambda_7$  are the numbers of internal  $D_1$  vertices on the paths and  $\mu_1, \dots, \mu_s$  are the cycle lengths, then

$$\lambda_i \geq 0, \quad \mu_j \in \{3, 5, 6, 7, 9, 10, 11, 12\}, \quad \sum_i \lambda_i + \sum_j \mu_j = 12. \quad (7)$$

There are exactly 166 unordered topologies in (7).

Subdivide every edge of  $J$  by its corresponding  $D_2$  vertex to obtain the fixed incidence skeleton. If a segment of  $R$  and a simple path in this skeleton have the same endpoints, their internal vertices lie respectively in  $D_1$  and in  $A \cup D_2$ . Their union is therefore a simple cycle, so the sum of their lengths cannot belong to  $\{4, 8, 16, 32\}$ .

The verifier enumerates every simple skeleton path and every locally legal path or cycle component. A component is represented by its set of  $D_2$  endpoint labels and its color-count vector. Exact set-packing dynamic programming then requires the endpoint sets to partition all fourteen labels and the color counts to equal (6). This search is a relaxation: it imposes only the one-segment necessary conditions, so an empty search space is sufficient for exclusion.

For the first two kernels, paths with respectively zero, one and two internal  $D_1$  vertices have 91, 8, 25 legal signatures; for the last two the counts are 91, 0, 20. Every longer path table and every cycle table is empty. The two nominal length-one paths in the first two kernels reuse the same endpoint pair, leaving only topology  $(0, 2, 2, 2, 2, 2, 2)$ . In that topology the union of the color supports of all legal length-two paths omits a color whose multiplicity in (6) is positive, for each of the four kernels. Thus no residual assignment exists.

The standard-library verifier `audit_a10_residual_gate_2026-08-03.py` independently regenerates the four kernels and the 166 topologies, enumerates the component signatures, and performs the exact packing. Its output status is `VERIFIED_COMPLETE_NO_ASSIGNMENT`; the canonical output and an independent replay are byte-identical with SHA-256

F3E39B881120A2B0FFDD7BCFE76E6778FDD2A38D2D3F66AE59745367E4767A25.

At the opposite end, equality is also impossible for  $a \leq 2$ . We use the following elementary finite lemma.

**Lemma 7.** *Every Hamiltonian graph on twelve vertices with minimum degree at least three contains a  $C_4$  or a  $C_8$ .*

*Proof.* Fix a Hamilton cycle and label its vertices alternately  $E_0, O_0, \dots, E_5, O_5$ , with subscripts modulo six. A chord at cyclic distance 3, 5, 7, or 9 immediately closes to a  $C_4$  or a  $C_8$ . Consequently every remaining chord joins vertices of the same parity. Since every vertex needs a chord, the chords in each parity class contain an edge cover of six vertices.

Choose an inclusion-minimal such cover. It is a star forest, so its component-order partition is 6, 4 + 2, 3 + 3, or 2 + 2 + 2. Comparing the two Hamilton-cycle arcs between pairs of retained chords eliminates the first three cases. Of the fifteen perfect matchings in the last case, exactly the following six survive within one parity class:

$$\begin{aligned} &01|23|45, \quad 01|25|34, \quad 03|12|45, \\ &03|14|25, \quad 05|12|34, \quad 05|14|23. \end{aligned}$$

Combining one surviving even matching with one surviving odd matching gives a  $C_4$  or a  $C_8$  in each of the 36 cases. The standard-library script `audit_hamilton12_lemma.py` independently enumerates the minimal edge covers and prints a literal cycle witness for all 36 pairs.  $\square$

Under equality, degree summation gives

$$n = 3a + 6, \quad |E(G)| = 5a + 9 + x/2,$$

where  $x \geq 0$  is even. For  $a = 0, 1, 2$ , the extremal thresholds are respectively 7, 12, 17, while  $G$  has at least 9, 14, 19 edges. The forced cycle divisible by four has length four at order six, length four or eight at order nine, and length four, eight, or twelve at order twelve. Only the last alternative survives, but it is Hamiltonian and Lemma 7 supplies a  $C_4$  or a  $C_8$ . Thus equality is impossible for  $a \leq 2$ .

We have proved the following strengthening.

**Theorem 8.** *Let  $G$  be a minimal counterexample, with  $a, d, n$  as above. If  $d = 2a + 6$ , then  $a \leq 9$ . Consequently, if  $n \geq 34$ , then*

$$d \geq 2a + 7, \quad d \geq \left\lceil \frac{2n + 7}{3} \right\rceil, \quad a \leq \left\lfloor \frac{n - 7}{3} \right\rfloor.$$

*Equality in Theorem 1 can survive only at the seven orders 15, 18, 21, 24, 27, 30, 33.*

*Proof.* The deficit analysis excludes equality for  $a \geq 11$ , the finite certificate excludes  $a = 10$ , and the small-end argument excludes  $a \leq 2$ . Hence equality implies  $3 \leq a \leq 9$ . If  $n \geq 34$  and equality held, then  $n = a + d = 3a + 6$  would force  $a \geq 10$ , a contradiction. Thus  $d \geq 2a + 7$ ; substituting  $a = n - d$  gives the two displayed integer bounds.  $\square$

## 6 A related size bound

The same minimality framework gives a useful accompanying estimate.

**Corollary 9.** *If  $m = |E(G)|$ , then  $m \leq 2n - 3$ .*

*Proof.* Choose a cubic vertex  $v$ . Every subgraph of  $G - v$  is a proper subgraph of  $G$ , so  $G - v$  is 2-degenerate. Hence

$$m - 3 = |E(G - v)| \leq 2(n - 1) - 3,$$

which gives  $m \leq 2n - 2$ .

If equality held,  $G$  would be an  $n$ -vertex,  $(2n - 2)$ -edge graph with no proper subgraph of minimum degree three. Narins, Pokrovskiy and Szabó [2, Theorem 1.4] proved that every such graph is pancyclic. In particular it contains a  $C_4$ , a contradiction.  $\square$

## 7 Verification and priority boundary

The proof was independently derived and then checked line by line by a separate agent. The kernel edge bounds were also exhaustively tested on every labelled graph through seven vertices; this computation is an audit aid, not a premise. The finite certificate for Theorem 8 was replayed independently from the exploratory solver, and its reduction and implementation received a separate hostile audit. Searches on 3 August 2026 for the exact constants and the incidence-kernel construction found Carr’s 4/7 theorem but no earlier statement of the bounds above. That is a search-relative novelty check, not a guarantee concerning unpublished or unindexed work. These theorems are necessary conditions on a hypothetical counterexample and do not resolve the Erdős–Gyárfás conjecture.

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## References

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